

Brief paper

Control loop performance assessment using detrended fluctuation analysis (DFA)[☆]B. Srinivasan, T. Spinner, R. Rengaswamy¹

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ABSTRACT

A new index is developed for assessing the performance of a single-input single-output (SISO) linear feedback control loop. The proposed metric is a specific scaling of the generalized Hurst exponent, computed through the method of detrended fluctuation analysis (DFA). We refer to this scaled exponent as the Hurst index. The new method compares favorably with the widely used minimum variance index (MVI), with both indices showing similar trends under changes in controller tunings during closed-loop simulations. The main advantage of the Hurst index over the MVI and other existing performance measures is that its determination does not require *a priori* knowledge of any loop parameters. Instead, computation of the index relies solely upon process output data collected during routine plant operation. Therefore, this new technique could potentially allow engineers to more efficiently identify problematic control loops.

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1. Introduction

Across the process industries up to 60% of control loops have less than acceptable levels of performance, with a corresponding deleterious effect on profits that could range into the hundreds of millions of dollars annually (Desborough & Miller, 2001). The first step to improving control loop performance is to determine which loops are behaving poorly (Jelali, 2006). Integral of squared error and output variance are commonly monitored measures of loop performance, but these statistics are essentially meaningless unless compared against a benchmark (Kozub, 1997). Harris (1989) has proposed benchmarking the mean squared error of the output against the theoretical minimum variance. This method uses knowledge of the process time delay along with routine operating data to find the controller invariant part of the process disturbance in order to calculate the minimum output variance possible under feedback control. This benchmark is based on the theoretical framework of minimum variance control previously developed (Åström, 1970; Box & Jenkins, 1970).

Controller performance assessment has been extended (Jelali, 2006) to the cases of multivariate and constrained controllers, and alternatives to the minimum variance benchmark have

been developed including the linear quadratic Gaussian and optimal achievable PID benchmarks. Among the various methods, the minimum-variance index has seen widest implementation in industry (Jelali, 2006) for several reasons. First, minimum variance control provides an absolute lower bound on achievable performance (Harris, 1989). Most importantly, its use requires only that a single parameter be specified *a priori*, that being the process time delay, whereas other methods require full plant and disturbance models. However, this requirement for *a priori* information outside of routine plant operating data is by no means negligible. First, a model of each process loop (or even its time delay) is not readily available in general (Jelali, 2006). Also, an existing process model may not include the true time delay, rather it could contain the delay that best fits the data to the widely used first order plus dead time (FOPTD) model (Björklund & Ljung, 2003). Second, determination of delay in most cases will require the control engineers to schedule an identification test with operating personnel. This is because routine plant operating data will not in general contain sufficient information for time delay determination, unless there have been abrupt control signal changes or external excitations during data collection (Jelali, 2006).

In view of the potential problems encountered in the calculation of time delay for minimum variance methods, it would be useful to have a performance measure whose theoretical backing is similarly strong to that of the minimum-variance index, yet requires no *a priori* knowledge of parameters for the control loop considered. Towards that end, we propose the use of the Hurst index, a metric based on the generalized Hurst exponent. Section 2 of this paper provides a review of the minimum variance benchmark, as well as a theoretical background for the generalized Hurst exponent and its determination through detrended fluctuation analysis.

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Section 3 contains a definition of the Hurst index and a discussion of connections to minimum variance control. In Sections 4 and 5, the results of a series of simulation and industrial case studies are presented, in which the behaviors of the Hurst and minimum variance indices are compared.

2. Preliminaries

In this section, an overview of single-input single-output (SISO) processes and the minimum variance index are presented, as well as details of the Hurst exponent, a quantitative measure for analyzing correlations in the time series data.

2.1. Single-input single-output system (SISO) process description

Consider the output Y_t of a discrete process that can be described as the combination of a linear process transfer function and a disturbance D_t given by $Y_t = \frac{B(q^{-1})}{A(q^{-1})}q^{-b}U_t + D_t$ where, as in Harris (1989), D_t has the form of an autoregressive integrated moving average (ARIMA) process with $D_t = \frac{\theta(q^{-1})}{\nabla^d \phi(q^{-1})}e(t)$ where q^{-1} is the backward shift operator; $B(q^{-1})$ and $A(q^{-1})$ represent vectors of parameters used to describe the linear transfer function; ∇ is $(1 - q^{-1})$; b denotes the integer number of delays between the controller output U_t and process output Y_t ; $\theta(q^{-1})$ and $\phi(q^{-1})$ are vectors of moving average and autoregressive coefficients used to model the disturbance D_t ; and $e(t)$ is an i.i.d. (independent and identically distributed) white noise sequence. The term ∇^d in the above expression is used to model the integrating nature of the corrupting noise.

2.2. Minimum variance index

For the above discussed closed loop system, the deviations from the setpoint $y_t = Y_t - Y_{sp}$ can be represented as (Harris, 1989) $y_t = (1 + \psi_1(q^{-1}) + \dots + \psi_{b-1}(q^{-(b-1)}) + \dots)e(t)$ where $\{\psi_j\}$ are impulse weights. It has been shown (Åström, 1970; Box & Jenkins, 1970; Harris, 1989) that when a minimum variance control law is applied to this closed loop system, the process output should theoretically be uncorrelated beyond a lag $(b - 1)$, due to the fact that the process output under minimum variance control can be modeled as an ARIMA(0, 0, $b - 1$) process. This leads to an expression for the theoretical minimum variance given by $\sigma_{MV}^2 = \sigma_e^2 \sum_{i=0}^{b-1} \psi_i^2$ where σ_{MV}^2 is minimum variance and σ_e^2 is the white noise variance. Using this idea, Harris (1989) designed the performance measure for SISO systems given by $\zeta(b) = \frac{\sigma_y^2}{\sigma_{MV}^2}$ where σ_y^2 represents the variance of the process output. When the value of the measure is 1, the loop is under minimum variance control, while values greater than 1 indicate the possibility of improving the performance of the control loop. Alternatively, the inverse of Harris' index, ζ^{-1} , takes on values in the range 0 to 1, and will be referred to as the minimum variance index (MVI). To obtain the theoretical minimum variance from operating data, the process output is fit with an ARMA (autoregressive moving average) model to determine the first b impulse parameters. Thus, computation of the MVI relies upon the process data and knowledge of the closed loop time delay. However, as mentioned in Section 1, the time delay is often both unknown *a priori* and not identifiable from the routine operating data themselves.

2.3. Generalized Hurst exponent

The generalized Hurst exponent is used in fractal analysis to quantify the presence of correlations in time series and has been applied in fields as varied as stock market analysis, dam construction, and bio-engineering. In fractal analysis, a time series

y_t is self-similar (contains sub-units which resemble the whole structure) if it holds that $y_t(k) \equiv a^\alpha y(k/a)$, where a represents the scaling factor along the x -axis (usually the time variable) and a^α the scaling factor along the y -axis. The exponent α is the self-similarity parameter. It is straightforward to see that if the scaling factors along the x and y axes are M_x and M_y , respectively, then the exponent is given by $\alpha = \ln M_y / \ln M_x$. By considering only the second order properties of the probability distribution functions of the scaled and original time series, it can be shown that for a zero mean process $\alpha = \ln \sigma_y / \ln \sigma_x$ (Palma, 2007). Here σ_y and σ_x are the standard deviations of the variables in the x (time variable) and y (magnitude variable) axes, respectively. The scaling exponent α for a correlated sequence is called the generalized Hurst exponent and can be obtained by mapping the time series to a self similar process through integration (Peng et al., 1992).

2.4. Computation of the generalized Hurst exponent using detrended fluctuation analysis (DFA)

DFA is one of the widely used methods to compute the generalized Hurst exponent (Hu, Ivanov, Chen, Carpena, & Stanley, 2001; Peng et al., 1992). The steps involved in DFA for computation of the exponent α are provided below (Hu et al., 2001; Peng et al., 1992).

- (1) Map the given time series y_t to the self similar process $Y(t')$ through removal of the mean and integration. In essence: $Y(t') = \sum_{k=1}^{t'} (y_t(k) - \langle y \rangle)$.
- (2) The integrated time series $Y(t')$ is segmented into W windows of length L . For each window j ($j = 1 \dots W$), a linear least squares fit is performed using the objective function $\min_{a_j, b_j} \epsilon(L)_j^2 = \sum_{i=1}^L (Y(i) - a_j i - b_j)^2$.
- (3) The root mean squared (rms) fluctuation is calculated as a function of window length using the formula

$$F(L) = \sqrt{\frac{1}{WL} \sum_{j=1}^W \epsilon(L)_j^2}.$$

- (4) Repeat steps 2–3, varying the window size between a minimum length of 10 samples and a maximum length of $N/4$ with N being the number of samples.
- (5) A log–log plot of the rms fluctuations against the corresponding window lengths should yield a straight line. The slope of this graph is the generalized Hurst exponent α .

2.5. Discussion on the computation of the Hurst exponent

In step 2 of the algorithm, the minimum window length for computation of DFA is chosen to be 10 while the maximum window size is chosen to be $N/4$. This selection is due to the reasons that (i) Hurst exponent calculation with window sizes greater than $N/4$ is generally unstable (Damouras, Chang, Sejdic, & Chau, 2010; Peng et al., 1992), and (ii) at least a minimum of 5–10 samples is required to compute the Hurst exponent. Recently, a rigorous statistical way to determine the minimum and maximum number of boxes was discussed in Damouras et al. (2010). However, in this work we used the general recommended settings for Hurst exponent calculation.

In step 3 of the above procedure, straight line fitting is performed to remove the linear trend present in the integrated data and is therefore known as DFA of order 1. This method is useful for analyzing the performance of regulatory control loops with routine steady state data. Simple modifications to step 3 of the algorithm are required to analyze the performance of (i) servo control loops where higher order polynomial fitting may be required depending upon the nature of trend introduced by the setpoint changes and (ii) control loops with oscillatory process output which introduces crossover or deviation from a

single straight line fit at the window size closest to the period of oscillation. This could potentially be used for oscillation detection in linear closed loop systems. It has been noted that the MVI is not recommended for oscillatory control loops (Ordys, Uduehi, & Johnson, 2007). A detailed discussion on computation of the Hurst exponent using DFA for data with various trends and oscillations is provided in Hu et al. (2001).

2.6. Generalized Hurst exponent and system behavior

The value of α for a stationary time series ranges between 0 and 1 with $\alpha > 0.5$ indicating the presence of long-term correlations in the data and $\alpha < 0.5$ indicating the presence of anticorrelation in the dataset. For a white noise sequence, the relation $\alpha = 0.5$ will hold. Other values of interest are $\alpha = 1$, which indicates the presence of pink noise ($1/f$ noise) (Bak, Tang, & Wiesenfeld, 1987), and $\alpha = 1.5$, which indicates the presence of Brownian noise (Palma, 2007).

Hurst exponent computation using the algorithm in the previous section was performed on the outputs of three simulated closed-loop systems. Samples of the output data generated for these systems along with plots showing their calculated autocorrelation functions (ACF) and rms fluctuations are presented in Fig. 1. Systems I–III have generalized Hurst exponent values of 0.93, 0.48, and 0.08, respectively. System I has an output that slowly varies between negative and positive values, with few zero crossings. As expected for a system with an α close to 1, its ACF plot indicates the presence of long range correlations. System II, with a generalized Hurst exponent relatively close to 0.5, appears less predictable and has autocorrelations which become insignificant/zero after a small number of lags. Row III of Fig. 1 illustrates the final step of the DFA procedure of Section 2.4. The linear behavior seen in the first two systems indicates the appropriateness of obtaining generalized Hurst exponent estimates through the use of best fit lines for these cases. The lattermost system, with a Hurst exponent closer to zero, appears oscillatory. Its ACF exhibits a decaying oscillation which also indicates periodicity as expected for a system with α close to zero. As discussed in Section 2.5, oscillatory time series such as the one corresponding to the third system will introduce crossover or significant deviation in the log–log plot of rms fluctuations versus the window length which can be observed from row III of Fig. 1 (here only first point deviates from a straight line fit). Such oscillation related issues are discussed in detail in Hu et al. (2001). A novel delay-free control loop performance measure for regulatory control loops developed using the generalized Hurst exponent is discussed next.

3. Connection between the MVI and the Hurst exponent—a new delay free measure

The MVI is $\frac{\sigma_{MV}^2}{\sigma_y^2}$ with $\sigma_{MV}^2 = \sum_{i=0}^{b-1} (\psi_i^2(q^{-i}))\sigma_e^2$ and theoretically $\sigma_y^2 = \sum_{i=0}^{\infty} (\psi_i^2(q^{-i}))\sigma_e^2$. If the output can be modeled beyond the closed loop lag of b , then the overall variance $\sigma_y^2 = \sum_{i=0}^{\infty} (\psi_i^2(q^{-i}))\sigma_e^2$ increases and the MVI as defined above goes below 1 indicating that better control can be achieved. For a linear system with minimum variance control, the process output can be modelled using ARIMA(0, 0, $b-1$), indicating that beyond lag b the process output should be white so that it cannot be predicted using a parametric model. In other words, predictability of the process output beyond lag b implies the possibility of better control for linear systems. This is also true for a restricted class of nonlinear systems as discussed in Harris and Yu (2007). The generalized Hurst exponent quantifies the amount of predictable/modelable

component present in the process output without using the knowledge of time delay b . An ARIMA(0, 0, b') process with autocorrelation function $C(s) \propto \exp(-s/s_x)$ (s_x being the rate of exponential decay) has a Generalized Hurst exponent value $\alpha = 0.5$, for small values of b' , as the number of samples $N \rightarrow \infty$ (for example, System II shown in Fig. 1) (Palma, 2007; Peng et al., 1992). It is clear, then, that the process output from a minimum variance control loop should have a generalized Hurst exponent value close to 0.5. Following these developments, a generalized Hurst exponent based performance measure which has a range of 0–1 similar to the MVI is defined as

$$\eta = \begin{cases} \alpha/0.5 & \text{if } \alpha \leq 0.5 \\ 1.5 - \alpha & \text{if } \alpha > 0.5 \end{cases} \quad (1)$$

where we will refer to η as the Hurst index. The process output from a loop under minimum variance control should have a value close to 1 while significant deviations from 1 indicate poor performance of the control loop.

3.1. Sensitivity of the MVI and the Hurst index to closed-loop time delay

For a system under minimum variance control, the process output follows an ARIMA(0, 0, $b-1$) model. As the time delay b increases, then the amount of predictable component in the output will increase, even if minimum variance control is maintained. Thus, the Hurst index, which measures the predictable component of the output without regard to fundamental system delays, will start to deviate from 1 for larger time delays. This will put the Hurst index at a disadvantage in systems where the time delay is known for certain, as it may indicate potential for improvement in feedback control loops where such potential does not exist. However, the following example demonstrates the efficacy of the new approach for systems in which the true time delay is not known with certainty.

Consider the set of systems, taken from Sendjaja and Kariwala (2009), with process and disturbance transfer functions $\frac{0.5108q^{-d}}{1-0.9604q^{-1}}$ and $\frac{0.5108}{1-0.9604q^{-1}}$, which differ only in their process time delays d . Simulated outputs were generated for a subset of these systems (having delays ranging from 1 to 15 sampling units) under minimum variance control. Computations of the Hurst index were performed for each system, along with a calculation of the MVI assuming a process delay of 3 sampling instances. Further, the MVI was also computed using the correct time delay information for each system, and all results are shown Fig. 2. Observing the figure, it appears that the MVI calculation is very sensitive to the assumed time delay information compared to the Hurst exponent based index. The index η varies from 1 to 0.8 for the range of different time delays, while the MVI varies from 1 to 0.3 if the wrong delay of 3 is chosen. The MVI calculation can be improved with better estimates of the time delay, but, as previously discussed, precise determination of the time delay requires external excitation of the control loop (Jelali, 2006). On the other hand, the Hurst index does not require any *a priori* knowledge of the process delay. Thus, the proposed measure is robust to changes in the closed-loop time delay and operates with the available routine plant operating data. Simulation studies performed on several systems taken from the literature are discussed in the following section.

4. Simulation studies

In Sendjaja and Kariwala (2009), optimal PI and PID controller parameters obtained through sum of squares programming are presented for several systems including one with the plant model $\frac{0.1q^{-3}}{1-0.8q^{-1}}$ and disturbance model $\frac{1}{1-q^{-1}}$. Considering this system separately for the cases of PI and PID control, variation of controller gain about the optimal parameter set was used to produce a

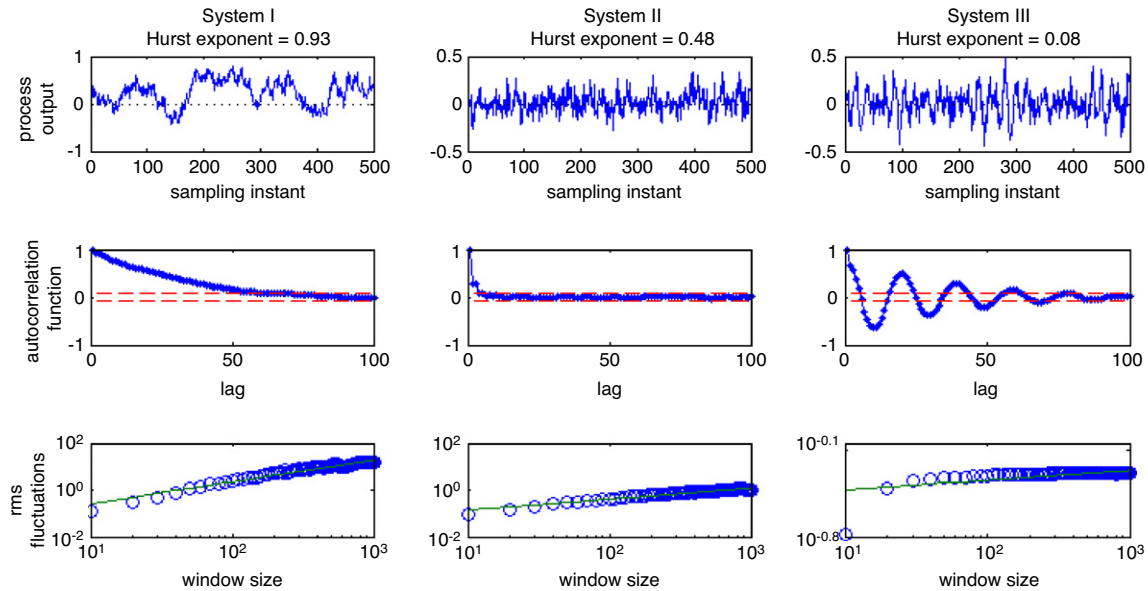


Fig. 1. Examples of systems with different Hurst exponents: row I—samples of output data from three processes; row II—ACF for the corresponding full data sets (4000 points) along with 95% confidence intervals; row III—log-plots of fluctuation intensity with best fit lines for the three processes.

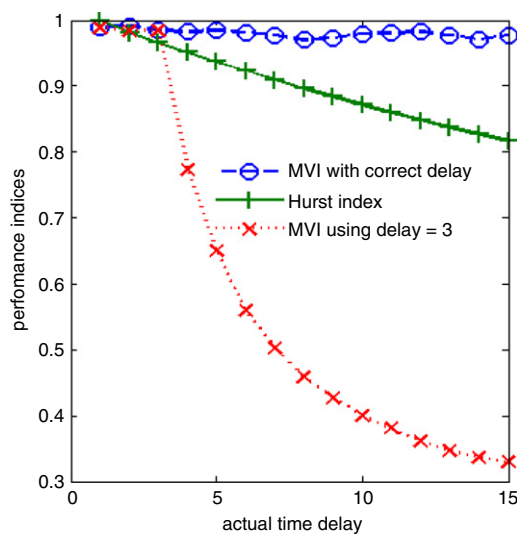


Fig. 2. Sensitivity of the minimum variance and Hurst exponent based indices to system time delay.

comparison of the Hurst index and the minimum variance index. At each set of parameters used, calculations of the Hurst index, MVI and integral of squared error (ISE) scaled with respect to maximum ISE value were performed (using 4000 data points) for each controller type. Minimizing the ISE is a common control loop objective, and its inclusion in this study allows for comparison of the Hurst index and ISE behaviors. The results from the study are shown in Fig. 3.

Notably, the overall trends of the minimum variance index and the Hurst index are consistent for each case studied. A second observation from Fig. 3 is that the Hurst index trend is related to the integral of square error (ISE) curve. This is already evident from the fact that the Hurst and minimum variance indices trend together and that the minimum variance index is, in a certain sense, a rescaling of the ISE; however, it is still useful to emphasize the Hurst index's correspondence with the ISE, which is an important metric for plant personnel. Significantly, the breakpoints (points where large changes in slope occur) in the curves of the two measures are aligned. Thus, the Hurst index compares favorably to both a

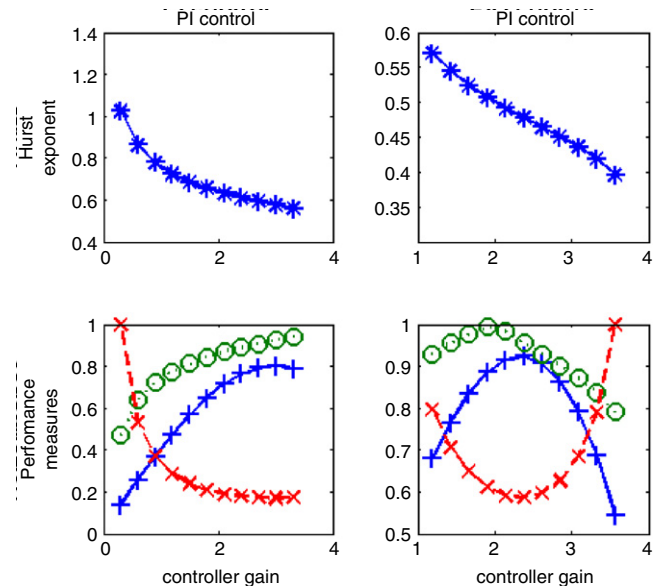


Fig. 3. Performance of a simulated linear system under varying controller gain: row I—plots of the unscaled generalized Hurst exponent; row II—co-plots of the Hurst index (+), minimum variance index (o), and scaled integral of square error (x).

popular performance metric (MVI) and a traditional plant objective (ISE) for a linear system controlled by PI and PID controllers.

Consider now the control loop presented in Doyle, Pearson, and Ogunnaike (2000), a jacketed CSTR (continuous stirred-tank reactor) with a polymerization reaction. The reactor model is nonlinear; however, the IMC controller that is implemented was designed through approximating the process using a second order Volterra model. Complete details of the plant model and controller are available in Doyle et al. (2000). The controlled variable is the number-average molecular weight and the manipulated variable is the flow rate of the initiator. Using the nonlinear plant model, IMC controller, and an additive output disturbance, the system was simulated for 3000 sampling instants for each of two disturbance types: (1) white noise and (2) white noise passing through a filter. A sample of output data for each case is shown in Fig. 4. The output in the case of colored noise distur-

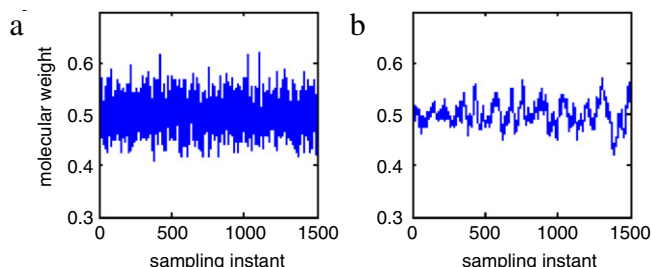


Fig. 4. Simulated output of the nonlinear polymerization plant model under IMC control with additive disturbance at the output of type (a) white noise and (b) colored noise.

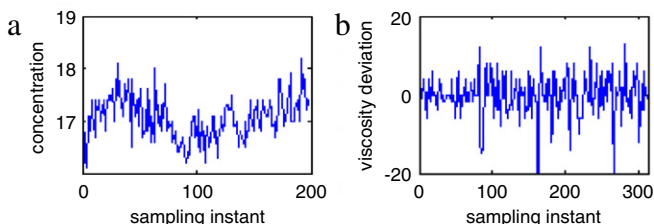


Fig. 5. Industrial datasets (Box & Jenkins, 1970). (a) Series A: “open loop” concentration readings and (b) series L: polymer melt viscosities from a closed loop process.

bance indicates long range correlation in the data, which a plot of its autocorrelation function (not shown) confirms. In the first case, use of the DFA procedure provides a Hurst exponent of 0.37, which corresponds to a Hurst index of 0.74. The deviation from 1 (indicating that control could be improved) can be explained by the use of an approximate plant model when developing the IMC controller. In the case of colored noise disturbance, a Hurst exponent of 0.90 and Hurst index of 0.60 are obtained. The IMC controller was designed for the purpose of set point tracking, and it is therefore unsurprising that it performs poorly at disturbance rejection.

5. Industrial case studies

Series A from Box and Jenkins (1970) (Fig. 5), contains 197 “uncontrolled” (compensated for the effects of the controller) concentration readings from a chemical process. It is stated in Box and Jenkins (1970) that the data can be fairly well fitted using a simple nonstationary model, so it is to be expected that the Hurst exponent will be much greater than 0.5. Indeed, through the DFA procedure, it is found that the Hurst exponent is 0.95, providing a Hurst index of 0.55. If the process time delay is assumed to be 1 sample, then the MVI is calculated to be 0.54, which agrees with the Hurst index.

Series L from Box and Jenkins (1970) (Fig. 5) has 312 observations of polymer melt viscosity, taken from a closed loop industrial process and presented in the form of deviations from the target value. Harris (1989) provides values of the standard deviation of the data (4.74) and the minimum variance for this process (4.08), which can be used to calculate an MVI of 0.74. Computation of the Hurst index through DFA provides a value of 0.79, which agrees well with the MVI.

6. Conclusions

A new control loop performance assessment measure is developed to identify performance degradation in closed-loop SISO systems. This measure uses the generalized Hurst exponent to indicate whether predictability is present in the process output. Unlike the well known minimum variance index for SISO loops, the proposed approach does not require any knowledge about the process time delay or other process parameters. Further,

the relationship among the proposed generalized Hurst exponent based measure, minimum variance index, and autocorrelation of process output has been discussed. Simulation and industrial case studies were performed to compare the results obtained from the proposed approach with the minimum variance index and integral squared error measures. The results obtained from the studies indicate the effectiveness of the proposed approach in monitoring the performance of closed-loop SISO systems.

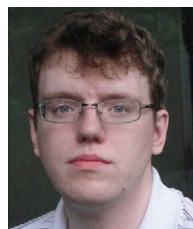
References

- Åström, K. J. (1970). *Introduction to stochastic control*. New York: Academic Press.
- Bak, P., Tang, C., & Wiesenfeld, K. (1987). Self-organized criticality: an explanation of $1/f$ noise. *Physical Review Letters*, 59, 381–384.
- Björklund, S., & Ljung, L. (2003). A review of time-delay estimation techniques. In *Proceedings of the 42nd IEEE conference on decision and control* (pp. 2502–2507).
- Box, G. E. P., & Jenkins, G. M. (1970). *Time series analysis: forecasting and control*. San Francisco: Holden-Day.
- Damouras, S., Chang, M. D., Sejdic, E., & Chau, T. (2010). An empirical examination of detrended fluctuation analysis for gait data. *Gait & Posture*, 31, 336–340.
- Desborough, L., & Miller, R. (2001). Increasing customer value of industrial control performance monitoring—Honeywell's experience. In *AICHE symposium series No. 326. Vol. 98* (pp. 169–189).
- Doyle, F. J., Pearson, R. K., & Ogunnaike, B. A. (2000). *Identification and control using volterra models*. London: Springer-Verlag.
- Harris, T. (1989). Assessment of closed loop performance. *The Canadian Journal of Chemical Engineering*, 67, 856–861.
- Harris, T. J., & Yu, W. (2007). Controller assessment for a class of nonlinear systems. *Journal of Process Control*, 17, 607–619.
- Hu, K., Ivanov, P. C., Chen, Z., Carpena, P., & Stanley, H. E. (2001). Effect of trend on detrended fluctuation analysis. *Physical Review E*, 64(1), 011114.
- Jelali, M. (2006). An overview of control performance assessment technology and industrial applications. *Control Engineering Practice*, 14, 441–466.
- Kozub, D. J. (1997). Controller performance monitoring and diagnosis: experiences and challenges. In *Proceedings of the chemical process control conference No. 316. Vol. 93* (pp. 83–96).
- Ordys, A., Uduehi, D., & Johnson, M. A. (Eds.). (2007). *Process control performance assessment: from theory to implementation*. Berlin: Springer.
- Palma, W. (Ed.). (2007). *Long-memory time series: theory and methods*. New Jersey: Wiley.
- Peng, C. K., Buldyrev, S. V., Goldberger, A. L., Halvin, S., Scortino, F., Simons, M., et al. (1992). Long range correlation in nucleotide sequences. *Letters to Nature*, 356, 168–170.
- Sendjaja, A. Y., & Kariwala, V. (2009). Achievable pid performance using sums of squares programming. *Journal of Process Control*, 19, 1061–1065.



microfluidic devices.

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